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Self-healing and optimal fault tolerant routing in wireless sensor networks using genetical swarm optimization

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ABSTRACT

A wireless sensor network (WSN) is used in area monitoring, surveillance, virtual reality, artificial intelligence, etc. The interaction between sensor nodes (SNs) and base station (BS) play a major role in WSN. SNs use more energy to transmit data, and the occurrence of faults in these nodes may lead to complete network failure. To ensure enhanced network performance, the incorporation of an efficient fault detection mechanism becomes essential in WSN. The major aim of the work is to improve the self-healing capacity of the network. To meet the above requirement, a fault-tolerant routing path identification with genetical swarm optimization (FTGSO) is introduced in this work. Genetical swarm optimization (GSO) is the integration of the two optimizations like Genetic Algorithm (GA) and Particle swarm optimization (PSO). The cluster head (CH) selection carried out on the basis of residual energy, coverage, communication cost and proximity. Further, a fault-free routing path is introduced by GSO, and a self-healing method is employed to resolve any network connectivity issues and resume the normal system operation. The analysis and simulation outcomes are compared with other optimization algorithms to verify the effectiveness. The evaluation results verify the enhanced performance of the introduced scheme with a higher PDR (packet delivery ratio) of 96.8%, lower energy consumption of 0.19 J with a minimum packet loss ratio of 3.2% and 30 ms of E2E (end-to-end delay) compared to other existing routing protocols. Also, the performance of the proposed method is compared on the basis of E2E, energy efficiency, (PLR) packet loss ratio and PDR.

1. Introduction

A wireless sensor network (WSN) involves various SNs (sensor nodes) used to sense environmental conditions and sensed data is given to the destination node. Wireless sensor network has discovered application in broad fields, including military observation, environmental monitoring, healthcare, forest fire identification, and so forth [1]. The WSNs made of small wireless sensing devices are outfitted with information processing and communication abilities. Making the connectivity coverage of the network is an important factor [2]. Connectivity is the process of confirmation of data that reaches the destination. WSNs offer several advantages over customary wired industrial observing and control frameworks, including expanded system scope, simple and quick establishment, versatility against single node failure and cost-effective maintenance [3]. The WSN is made out of numerous sensor nodes, where sensor nodes are associated with different sensors using wireless communication channels.

Sensor nodes (SNs) have battery and less amount of energy and their aim is to enhance the lifetime of network in WSNs. One of the way to prolong the lifetime of network is the formation of heterogenous WSN model with super-nodes and it is a another way to monitor the center of WSNs [4]. Heterogenous WSN model with super-nodes has more network lifetime and reliable than the homogenous network. Hence, in this work heterogenous WSN model is considered. In WSN, the packet loss is high due to less lifetime of SNs. When the node dies in the path to transmit packet from source to destination, packets will loss because of unsuccessful transmission. The network must be able to rectify the fault of node and identify the better solution for better transmission [5].

Coverage is one of the most essential criteria expected in the WSN in recent times to enhance the performance of the network and to enhance its reliability [6]. It demonstrates the capability of the nodes in the network to monitor a particular field of interest to gather more information. Sensing ability of the sensor nodes deployed within the network is required to be focused to enhance the communication between the

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nodes so as to execute the applications faster. Based on the coverage criteria, the scheduling of sensor nodes is carried out to result in better performance [7, 8]. Another major issue arising in the WSN is the formation of holes due to the deactivation of sensor nodes in the network. Since the sensor nodes are limited in battery power, the energy gets depleted easily when there is a rush in transmission of data packets. When, the sensor node in a particular area gets depleted, then the node gets deactivated leading to hole formation [9, 10]. Mostly, the regions in the network with limited coverage or the regions that are unable to be sensed by the sensor nodes are highly prone to hole formation. In certain applications, to avoid hole formation, enable the deployment of adequate sensors in the network that is sufficient for covering the entire network [11].

Security in WSN is another major factor that is required to be evaluated to prove the reputation and reliability of the network. The data that is being transmitted within the network consists of both sensitive and insensitive information. Among them, the sensitive data is required to be secured as the privacy of the user is required to be protected. In the WSN scenario, the security of the data can be provided by adding an encryption methodology where the decryption is carried out only on the receiver's side [12, 13]. Lightweight authentication is of utmost significance in today's world to protect the data as well as to maintain network performance. Recently, there are several methodologies developed to protect the data and most of the methods integrated the data aggregation with security to attain reliability and higher performance [14].

In WSN, because of the less resources of SNs and unintended communication like water, snow, wind and rain the fault occurs. Hence, enhancing fault tolerance in network will enhance the quality of service (QoS) and accessibility of WSN. Further, addressing fault tolerance problem has been a necessary need in the design of WSN. The clustering process groups the SN nodes into small clusters and every clusters hacve cluster head (CH). Data packets are send to the base station (BS) from SN through CH. There are two types of communications; inter-cluster and intra cluter. In intra cluter communication, they energy of CH near BS may lose energy because of more data packets. Hence, the CH may die and cause failure in nodes. To tackle these issues, this work introduces an optimization based fault tolerant model. For the better WSN applications, the factors like high throughput, less energy consumption and self-healing in faulty nodes should be satisfied. Hence, to achieve this the hybrid optimization fault-tolerant routing path identification with genetical swarm optimization (FTGSO) is introduced.

The contribution of the proposed work is to develop a fault free routing model for WSN and to integrate this with multi-objective analysis for selecting the best node as CH. The proposed system has more number of SNs and BS. A fault-free routing path is introduced from the sensor node to the base station with GSO to avoid network failure. PSO and GA are GSO's methods and help to avoid failure in the network and enhance the performance of WSN. The main contributions of this research are listed below:

a Introducing swarm intelligence to identify fault free routing path in WSN, which integrates the properties of PSO and GA effectively for data forwarding.
b Identifying the efficient cluster head through multi-objective for data forwarding. The different multi-objectives considered are communication cost, residual energy, coverage, and proximity.
c Introduces PSO technique for faulty node detection in the cluster and GA operators to determine the fault free route. Both PSO-GA methods are used for enhancing the network lifetime.
d The overall efficiency of the network is improved by integrating the self-healing method. Self-healing solves the problem of network connectivity and packet collision.

The remaining part of the work is: Section 2 summarises the literature survey conducted to examine other approaches for fault-tolerant routing. Section 3 covers details of the methodology of the proposed system, and Section 4 provides the evaluated results of the work. Finally, the paper is concluded in Section 5.

2. Related works

WSNs have a wide range of applications, and a huge amount of research has contributed to the field. But, implementing and executing a fault free WSN application is challenging due to varied factors. The sensor nodes deliver surprising reactions due to asset limitations that prompt different sorts of faults. WSN framework is affected due to the presence of faulty SNs. Swain et al. [15] proposed regression-based fault detection and tolerance strategy for WSN. With the help of information from fault-free SNs, fault detection is carried out. Only a smaller number of sensors were accessible in those frameworks, and the fault tolerance module was distributed and it was incorporated with the base station.

In the innovation of WSNs, the fault diagnosis in view of fusion data analysis has attained considerations in the field of WSN. He et al. [16] depicted the data characteristic of WSN and a strategy for fault diagnosis of WSN in view of a belief rule base (BRB) model. A covariance matrix adaptation evolution procedure algorithm was utilized to improve the parameters. Alternate fields of WSN such as reliability evaluation, network security detection and data fusion were not considered with BRB strategies.

Gao et al. [17] presented a distributed filtering model to manage the issue of fault detection of nonlinear stochastic frameworks with WSN. Also, a fault reference demonstration was embraced to enhance the execution of the fault detection system. Another general fault detection framework was figured in an IT2 (interval type-2) T-S (Takagi–Sugeno) fuzzy model system. The framework was considered only for linear systems, and nonlinear frameworks with complex correspondence were not considered.

The UWSN is comprised of different UW-sinks arranged amidst regions under surveillance. Goyal et al. [18] determined FDRT (fault detection and recovery technique) for clustering. Ack up cluster head (BCH) is chosen to utilize the fuzzy procedure considering parameters, for example, quality of a link, sink separation, the problem of load, the value of residual energy and density of node. During the fault detection at CH, the backup cluster head will identify the fault detected CH. When the BCH fails, another CH is selected as BCH, and it goes on. The FDRT delivery ratio is found to have a smaller number of faults.

WSN is comprised of gadgets that are disseminated inside the topographically segregated regions. Energy utilization is an essential concern for WSN. Elshrkawey et al. [19] introduced a method called TDMA (Time Division Multiple Access) for increasing the lifetime of WSN and reducing the energy consumption of the network. The strategy was proficient by expanding the energy adjustment inside the clustering to reduce the energy dissipation level and used for increasing the lifetime, CH number, and packet number given to the BS contrasted with the LEACH protocols. Indisputably, the exploration had been completed just to diminish the energy preservation and system lifetime, yet it doesn't exhibit any security affirmation to the WSN.

Imbalance in the consumption of energy while the data packets between the nodes is WSN is one of the major drawback. Due to the limited battery power of sensor nodes, it is important to consider the overall energy consumption in transmission. To address the issue, Shahbaz et al. [20] presented a multipath routing methodology that enabled load balancing in the network with enhanced scalability. The method involved three phases such as clustering, path discovery and path maintenance. The firefly algorithm was utilized for clustering and the fuzzy logic was adopted for discovering the path between CHs. The method outperformed the existing methodologies in terms of different metrics. Network traffic causes a lot of problems in data transmissions and a proper routing protocol to deal with this issue of utmost importance. Hatamian et al. [21] presented a congestion aware routing protocol that tried to find the most affordable routes in the network for

transmissions. The routing was performed using the greedy approach and the rate was controlled using a fuzzy rate controller. Congestion score and buffer occupancy were the criteria provided as the inputs for the controller and the results demonstrated the efficacy of the methodology. The main reason behind energy consumption is the transmission of data packets. One effective way to solve this issue is the data aggregation technique that decreases the number of transmitted packets. Naghibi and Barati [22] presented a methodology for data aggregation in WSN by combining the related data and preventing sending of additional packets. A secure hybrid structure data aggregation (SHSDA) is proposed in that work that adopted the tree and star structures for the implementation. The lightweight symmetric encryption was utilized to protect the data and a stable star structure was followed for network representation. The methodology reduced the delay in data delivery compared to other techniques.

Another secure data aggregation technique was built by Yousefpoor et al. [23] to minimize the data transmissions on large-scale WSNs. The technique followed intra-cluster data aggregation using the fuzzy scheduling system, inter-cluster data aggregation using an aggregation tree where, the dragonfly algorithm (DA) was adopted to find the optimal tree between CHs. A symmetric and lightweight encryption model based on residue number system (RNS) was utilized to enhance security in the network. An integration of hierarchical routing and data aggregation was introduced by Sharifi and Barati [24] to enhance the overall network lifetime. The centrality of the nodes, residual energy and distance between sink and node were identified and the fitness was defined to cluster the nodes present in the network. Then, the tree structure was formed and routing was performed based on the sink and source node locations. The evaluations of the model proved that the method expanded the number of alive nodes and packet delivery ratio. A tree-based data aggregation scheme was introduced by Hasheminejad and Barati [24] that followed the binary tree representation to demonstrate the organization of sensor nodes. Based on the authentication of a shared key, the aggregation process was initiated. For every hop, the error generated along the path was detected and whenever an error was detected in the network, then a retransmission request was sent to the previous hop. The lifetime of the network was enhanced in the method based on the reduction in energy consumption and the count of sent packets. Finally, evaluations proved the efficacy of the method in terms of reliability, network lifetime and energy consumption.

An energy efficient layered routing protocol (EELRP) was developed by Hajipour and Barati [26] to enhance the network lifetime and to support reliable communication. The network was initially divided into concentric circles where each circle was partitioned into 8 equal sectors. The nodes were then placed within each sector and the most appropriate one was elected as the agent. The sensed data within each sector was sent to the agent which aggregated the data, detected and corrected the errors using the redundant RNS system. The data was then traversed to the lower section of the same sector and the process was repeated until the data reached the base station. Another energy-efficient method for hole repair in WSN was proposed by Dezfouli and Barati [27] to achieve effective coverage. Maintaining a trade-off among energy consumption, functional coverage and network lifetime is challenging and the method resolved the issue with the use of mobile nodes and firefly algorithm. The network was initially partitioned into smaller equal cells to facilitate the prediction of physical coverage. Operational scheduling was followed to deactivate the nodes after evaluation of the network to ensure coverage. The optimum deployment of mobile nodes was ensured using the firefly algorithm to enhance the coverage under a circumstance of insufficient coverage in the cells. The evaluations of the method proved the efficacy against the compared methods.

Table 1 depicts the challenges and merits of the existing models. Several fault tolerance algorithms are available in the literature, defined at various levels. The faults must be identified and eliminated from the network at the correct time to avoid disasters. Existing techniques detect network faults and do not provide any recovery algorithm. Further, these models efficiently detect the faults. They achieved poor performance based on security, PDR and accuracy. Therefore, this work introduces the optimization method for fault detection and its rectification. Two types of networks used in WSN are heterogeneous WSN (H-WSN) and homogeneous WSN, and uniform nodes are in the

Table 1
Challenges and merits of the existing models.

Authors and citation	Methods	Merits	Challenges
Swain et al. [15]	Regressional-learning	This model achieved a better R^2 value of 0.99	Predictor sensor nodes were regarded as fault-free in nature
He et al. [16]	BRB	Achieved more effective evaluation of WSN fault diagnosis	BRB in WSN was not popular
Gao et al. [17]	IT2- T-S	Successfully provided simulated trailer model	Need improvement in FAR (False Alarm ratio) and detection accuracy
Goyal et al. [18]	FDRT	Fast failure detection and lifespan were improved	This model showed poor performance on the delivery ratio
Elshrkawey et al. [19]	TDMA	Energy consumption was less, and lifetime was increased	Security is limited
Shahbaz et al. [20]	Firefly algorithm, fuzzy logic	Reduced energy consumption, achieved effective load balancing with enhanced scalability	Costly as the sensors are equipped with GPS
Hatamian et al. [21]	Greedy approach, fuzzy logic	Transmission was carried out based on priority to reduce the congestion caused by upstream traffic	The quality of the data being prioritized is under question
Naghibi and Barati [22]	SHSDA	Reduced the number of transmitted packets in the network	Require more attention in aggregation while adding or removing nodes from the network
Yousefpoor et al. [23]	Fuzzy scheduling system, DA, lightweight symmetric RNS	Minimized the traffic effectively in the large-scale WSNs	Require more attention in the security of the system
Sharifi and Barati [24]	Data aggregation and hierarchical routing	Reduced the latency in data delivery abruptly	Efficiency and computational cost are required to be analyzed
Hasheminejad and Barati [25]	Tree-based data aggregation	The reliability of the entire network was enhanced with the reduction in sent packets and energy consumption	Due to the error detection procedure in every hop, the end-to-end delay of the network was increased
Hajipour and Barati [26]	EELRP	Effectively transmitted the data between the nodes and enhanced the network lifetime	The dynamicity of the network is not handled
Dezfouli and Barati [27]	Firefly algorithm	Enhanced the physical coverage in the network effectively	The increase and decrease of nodes in the network is required to be focused in terms of coverage

homogenous WSN. The important drawback of homogenous WSN is role rotation. Hence, the entire nodes in the network act as CH, and they possess the capabilities of hardware requirements. In the earlier stage of WSN, static nodes are used, and the communication module inside the static node consumes more energy. Also, most of the existing frameworks are solely formulated only to extend the lifetime of the network whereas, the faults arising within the network are not addressed. Thus, the proposed work is formulated to provide fault tolerance where the fault free routing path is determined for data transmissions. Moreover, the lifetime of the network is also enhanced with efficient transmissions.

3. Proposed methodology

WSN consists of a large amount of SNs, and they are utilized for computation purposes, sensing data etc. These SNs communicate with each other and communicate directly with BS. Radio trans-receiver, electronic circuit (for interfacing with sensor), microcontroller and battery are the components of a sensor node. Node is also known as a mote, which is used for monitoring the data, process the data, and used for communicating with other connected nodes. The topology of WSN uses either a simple star or multi-hop wireless model. In the simple star topology, every host is and central hub are connected, and a multi-hop wireless network organizes the nodes in a meshed manner. Routing and flooding are the two types of propagation techniques involved in WSN. Health care monitoring, area monitoring, water quality monitoring, surveillance and natural disaster monitoring etc., are some of the applications of WSN.

Fig. 1 represents the architecture of WSN. It is composed of sensor nodes, cluster head, sink node and base station (BS). SNs provide excellent performance because of the extensive range of resource constraints and critical applications, which causes different types of SNs faulty and generate problems in WSN. The decision-making process and overall accuracy of WSN are affected by the faulty sensor nodes, which causes the complete network system failure. Due to these problems, detection of fault is necessary. H-WSN has different kinds of nodes. Due to the overload problem, the CH become dead before other SNs. H-WSN reduces the hardware cost because a large number of hardware and the additional battery can be implemented in the CH node of H-WSN.

Unattended deployment, faulty SNs and a hospitable environment are the major factors responsible for failure in WSN. Network failure and SN failure are the two types of faults in WSN. In network failure, faults and errors occur inside the network area, and these problems are related to the environment and connectivity of the network. Due to the volatile nature of WSN, nodes are not aware of their surroundings and make faults during the data transmission. Due to the errors in hardware (low level), nodes are crashed and cause faults in SN. This is known as SN faults. There are lots of methods for fault detection in WSN, and the traditional method uses an agent packet transmission scheme.

To enhance the performance of WSN, the fault-tolerant genetical swarm optimization (FTGSO) method was used. In this work, heterogeneous WSN architecture is used. H-WSN consists of sensor nodes with different abilities (sensing range, computational power etc.). Heterogeneity is a function of energy, and in H-WSN, SNs are formed into clusters, and each SN has CH. The proposed model helps to find out fault nodes and avoid connectivity issues.

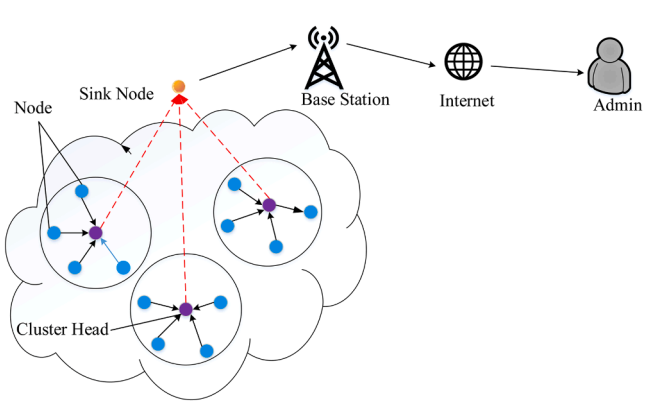

Fig. 1. WSN Architecture.

3.1. Network model assumptions

This section presents details of the network model consideration that are followed in the proposed work.

1 The whole network is considered static. It means all the network nodes are deployed in a stationary manner.
2 Nodes employed in the network are heterogeneous in nature. Normal, intermediate, and advanced are the three categories of heterogeneity. Transmitting and receiving of data in their communication area are the same and fixed for all the nodes.
3 The sink node is placed outside the network, and nodes inside the cluster consume more energy.
4 Nodes cannot be recharged again if they exhaust their energy, and these nodes are known as dead nodes.
5 Wireless is the communication medium. Hence reflection, refraction, attenuation of the signal, interferences etc., do not affect the overall operation. Packet loss and hindrance during communication are considered negligible.
6 GPS is not installed in the location of nodes.
7 For monitoring purposes, the square shaped area is used.
8 To initialize the communication process, sink nodes send a Hello message to the network area, and each node transmits its unique ID to the sink node. The Euclidean distance method is used to estimate the distance between nodes and the received signal strength indicator (RSSI) value used for the computation process. The collected unique ID is broadcast to the network area to make knowledge about neighbor nodes.
9 Unique ID consists of energy values and distance between node and sink.
10 In the proposed work, data is compressed, and it is considered immune to the noise.

3.1.1. Network model formation

In this section, the formation of the network model is described.

Normal (least energy), intermediate (middle level energy) and advanced (maximum energy) are the three types of heterogeneous nodes. The total number of normal nodes, intermediate and advanced, is denoted byN_{nrm},N_{int},N_{adv}. A portion of intermediate and advanced nodes out of a total number of nodes is represented bym_i and m_a.

$$N_{nrm} = n_o \times (1 - m_a - m_i) \tag{1}$$

$$N_{\text{int}} = n_o \times m_i \tag{2}$$

$$N_{adv} = n_o \times m_a \tag{3}$$

where n_o is the number of nodes

Nodes with a factor of percentage population of an intermediate and advanced node isγ and β. The energy of normal, intermediate and advanced nodes are represented byE_{nrm}, E_{int} and E_{adv}. The energy due to the normal node is given by

$$E_{nrm} = n_o E_e (1 - m_a - m_i) \tag{4}$$

wher $E_e$is the initial energy.

The energy due to the intermediate node is given by

$$E_{\text{int}} = n_o E_e \times (1 + \gamma) m_i \tag{5}$$

The energy due to the advanced node is given by

$$E_{adv} = n_o E_e \times (1+\beta) m_a \tag{6}$$

The energy available with the entire 3 nodes is given in Eqs. (4)-(7). E_{total} gives the total energy of the network and is estimated as follows.

$$E_{total} = E_{nrm} + E_{\text{int}} + E_{adv} \tag{7}$$

$$E_{total} = E_e(1 - m_a - m_i)n + E_e(1+\gamma)n_o m_i + E_e(1+\beta)n_o m_a \tag{8}$$

$$E_{total} = n_0 E_e - n_0 m_i E_e - n_0 m_a E_e + n_0 m_i E_e + n_0 \gamma m_i E_e + n_0 m_a E_e + n_0 m_a \beta E_e \tag{9}$$

From Eq. (9), it is seen that terms like second and fourth are equal in magnitude but out of face. Similarly, the terms like third and sixth are equal in magnitude but out of face. Therefore, Eq. (9) can be written as:

$$E_{total} = n_0 E_e n_0 \gamma m_i E_e + n_0 m_a \beta E_e \tag{10}$$

$$E_{total} = n_o E_e (1 + \gamma m_i + m_a \beta) \tag{11}$$

From Eq. (11), it is concluded that when the level of heterogeneity is added to level 2, the energy is improved by $1 + \gamma m_i$, and when it is improved to level 3, energy is improved by$(1 + \gamma m_i + m_a \beta)$.

3.2. Fault tolerant genetical swarm optimization

Swarm intelligence-based fault detection and recovery method are used in WSN by consolidating the aspect of energy efficiency. With the help of the fault detection algorithm, the working status of SNs can be monitored. The sensor node fault should be identified before the data gets processed to assure the system's decision correctness. A multi-objective fault tolerant genetic swarm optimization (FTGSO) algorithm is proposed in this work. Residual energy, coverage, communication cost and proximity are considered multiple objectives. In this FTGSO, the transmission of data by optimal routing path and self-healing is used to rectify of the problem of network connectivity.

Fig. 2 represents the proposed architecture model that consists of BS, SNs (with and without fault), and CH (with and without fault). CHs are selected on the basis of multi-objectives in terms of communication cost, proximity, residual energy and coverage. CHs are connected directly to BS. The node which ensures these objectives in an effective manner is considered CH. After the selection of CH from clusters, the best routing path is chosen for data transmission. If there is any fault occurs in the forwarding path, then the fault tolerance algorithm is applied with the help of genetic swarm intelligence, and a self-healing approach is used for addressing any connectivity issue of the network.

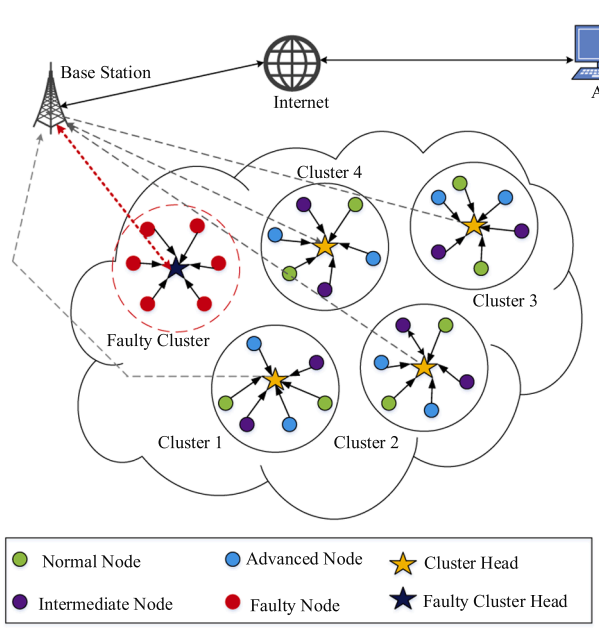

Fig. 2. Proposed Architecture Model.

3.2.1. Energy model

This framework is composed of radio energy dissipation and multi-path fading. Transmitter gives energy for radio electronics and power amplifiers. By that time, the energy is deliver to radio electronics using receiver. Distance between receiver and sender is denoted by D. If$D$is larger than the threshold value, then the free space model gives energy dissipation asD^2, and the multipath model givesD^4For transmission of$j$ packets, energy consumption$E_c$is denoted by

$$E_c = \begin{cases} j^*(E_C + E_f{}^*D^2) \; ; & D < D_0 \\ j^*(E_C + E_m{}^*D^4) \; ; & D \geq D_0 \end{cases} \tag{12}$$

$E_c$indicates energy for simulating SN electronic circuits. The distance between the transmitter, receptor and tolerant bit error is used for identifying the amplifier energy of multipath fading $(E_m{}^*D^4)$ or free space $(E_f{}^*D^2)$. E_f andE_m denote transmitter and amplifier model of system.$d_t$is the distance of threshold and represented in Eq. (11).

$$d_t = \sqrt{\frac{E_f}{E_m}} \tag{13}$$

Eq. (12) denotes the amount of energy consumed for packet reception.

$$E_{recp} = j^* E_c \tag{14}$$

The amount of data aggregation in CH is represented by Eq. (13).

$$E_{ch} = E_{ag}{}^* j^* m \tag{15}$$

where E_{ag}represents a value of energy needed for one-bit aggregation and$m$is the number of messages.

3.2.2. Genetical swarm optimization (GSO)

The genetical swarm optimization method is composed of GA and PSO. GSO helps to overcome the combinatorial optimization problem and maximize the lifespan of WSN.

3.2.2.1. Overview of PSO. PSO is an evolutionary-based optimization method that resembles the nature of flocking birds [29]. The population of PSO is known as a swarm. Each particle moves into the problem area with the help of the swarm's best position. The velocity value of the previous position is written as the swarm best position. The velocity and position of each particle are denoted as:

$$ve_{in}^{x+1} = ve_{in}^{x} + h_1 k_1^x \left[y_{in}^x - p_{in}^x\right] + h_2 k_2^x \left[m_n^x - p_{in}^x\right] \tag{16}$$

$$p_{in}^{x+1} = p_{in}^{x} + ve_{in}^{x} \tag{17}$$

where p_{in}^x and $ve_{in}^x$are the position and velocity of each particle in x dimension at inparticle, y_{in}^x and m_n^x are the previous best and global best solution in x dimension and h_1 , h_2 are cognitive and social constant (used for velocity control). Randomly distributed variables are denoted byk_1 and k_2.

3.2.2.2. Overview of GA. For producing a high-quality solution, GA is preferred [28]. Based on biological operators (selection, crossover, and mutation), GA solves the problem of optimization and search. The first stage of GA is the generation of chromosomes. Each chromosome in the GA indicates the geographical placement of sensor nodes within the network. The fitness value of the individual is estimated with the help of the fitness value of the given problem. High fitness values indicate more probability of survival. The second stage of GA is crossover. In this section, two parents cross (exchange) genetic traits. The mutation is the

final stage of GA that is used to increase offspring diversity, and the value of the gene is swapped. The goal of any generation is that the average fitness value of the population must be higher than the previous one. GA eliminates the satisfactory level of fitness value and maximum value of epochs.

The general drawback of GSO is it does not consider multi objectives at a time. Hence, the proposed model introduces multi-objective genetical swarm optimization (MO-GSO) for optimal fault tolerant routing in WSN. This work is about fault detection and recovery of WSNs by consolidating the aspect of energy efficiency and swarm intelligence-based routing. The working status of the sensor node is monitored by a fault detection algorithm which divides the failed sensor node in time or changes the equivalent module to assure that the failed node does not affect the function of the SN. The sensor node fault is found before the data is processed to assure the system's decision correctness.

Due to scalability, and longer lifetime, the unequal node clustering method is used. This method reduces the overload problem. BS send message to the network after the placement of nodes. SN gives information about their location after the reception of the message. Further, the lower and upper range of the network is analyzed. The lower range is estimated by calculating SN energy utilization and the upper range find out by using SN density.

With the help of multiple objectives (coverage, cost of communication, residual energy, and proximity), CH is selected from nodes. CH uses more energy than other nodes in the cluster. This energy is utilized to transmit and receive data from other SNs and for data aggregation. Other node uses energy for data collection, the transmission of data, and the reception of data. Hence CH is selected based on a multi-objective process. In this MO-GSO, data transmission is done by optimal routing path, and the performances are monitored.

3.3. Mathematical model and genetical swarm optimization algorithm

3.3.1. Cluster formation

Initially, assuming every node inside the cluster must have a valid routing path and BS generates a list$L_i$that includes every one hop CHs for CH. One hop CHs denoted by$a$and$b$is used for indicating CH. To satisfy the above condition, there is a communication link $b \rightarrow a \forall a \in L_i$ between CH and BS. For designing possible routing paths, these instructions are considered. Next, CH is used for the selection of the initial swarm formation. i.e.$a \in L_i$ for every$b$.

3.3.2. Selection of CH

A probabilistic model is used to select CH, which selects the best node as CH in a cluster on the basis of multi-objectives (proximity, cost of communication, residual energy value and node coverage) analysis. The energy is utilized by a node in the network for reception, collecting data and transmission processes. The energy is utilized in CH nodes more than in the other nodes in the clustered network due to data transmission and reception from several sensor nodes. A large amount of energy is required for the nodes to perform this process. A robust CH selection is needed for this, and to enhance the data aggregation efficiency, multi-objectives are used for selecting the stable CHs. Fig. 3 shows the architecture of multi-Objective Based CH Selection.

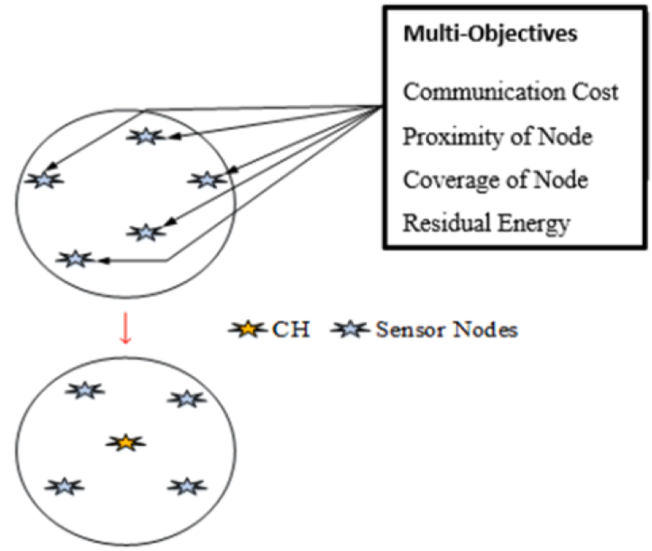

Fig. 3. Multi-Objective Based CH Selection.

3.3.2.2. Obj 2: Cost of communication. Cost used to communicate with the nearest node and calculated using Eq. (18).

$$C_C = \frac{dis_{avg}^2}{d_r^2} \tag{20}$$

where dis_{avg}^2represents the node average distance and$d_r^2$is the neighbor average distance. The second objective is shown in Eq. (19).

$$\min F_{f2} = \frac{1}{SN_t}\sum_{y=1}^{X} C_c(X_y) \tag{21}$$

3.3.2.3. Obj 3: Node residual energy. Node residual energy is calculated by Eq. (20).

$$R_e = E_{sn} - (E_{co} + E_{trans} + E_{recp} + E_{ag}) \tag{22}$$

where E_{sn} represents SN total energy and$E_{co}$is the energy used for data collection. Energy used for data transmission is denoted byE_{trans}, and the third objective is indicated by Eq. (21).

$$Max\ F_{f3} = \frac{1}{SN_t}\sum_{y=1}^{X} R_e(X_y) \tag{23}$$

3.3.2.4. Obj 4: Node coverage. Node coverage is estimated by Eq. (22)

$$N_c = R(X_y) \tag{24}$$

where$R(X_y)$ are the node radius and fourth objective represented in Eq. (23).

$$\max\ F_{f4} = \frac{1}{SN_t}\sum_{y=1}^{X} N_c(X_y) \tag{25}$$

A node which satisfies all 4 objectives is considered CH. That is, nodes with more node coverage and energy at less proximity and low cost is chosen. The CH obtained in each cluster is accountable for aggregation and packets are moved directly or indirectly to BS. The CH is selected, the route is determined to transmit aggregated data to BS.

3.3.2.1. Obj 1: Proximity of node. Node proximity is calculated by using node distance and given by Eq. (16).

$$N_P = \frac{1}{SN_t}\sum_{y=1}^{1-SN_t} N_n(x,y) \tag{18}$$

SN_t indicates the number of SN and$N_n(x,y)$is the distance between nodes, and the first objective is shown by Eq. (17).

$$\min F_{f1} = \frac{1}{SN_t}\sum_{y=1}^{X} N_p(X_y) \tag{19}$$

3.3.3. Selection of routing path

The maximum lifetime of WSN is used to compute the routing path. The maximum lifetime is calculated by

$$N_L = \frac{R_{initial}}{R_{\max}} \tag{26}$$

where$R_{initial}$ denotes the initial energy of CH and $R_{\max}$indicates the maximum amount of energy used by CH for a single round process.

3.3.4. Fault detection of node

There are several reasons for the WSN's failure. Initially, SNs fail because of battery depletion and damage due to external events. SN transmit data to the CH if no fault occurs. To discover the faulty nodes for enhancing the efficiency on the basis of successful PDR, energy consumption and energy efficiency. During the diagnosis of the fault, each node inside the cluster transmits data to the nodes of each CH. The lowest fitness value of the routing path from overall clusters is taken as a faulty cluster. Nodes inside the faulty cluster are considered a faulty node, and CH is identified as a faulty CH. Notaion and meaning of the symbols are given in Table 2.

3.3.5. Fault tolerance process

Network connectivity and partitioning problem arise due to faults in SN. A new position is obtained for each CH node during the fault tolerance stage. To provide fault free path for data transmission, CH ID and replacement ID are used. CH is taken as$b$ then list of their connected nodes are denoted byL_b CH replacement for fault-free path explained below

Example:$L_1 = \{3\ ,\ 5\}$ is the neighbor list of CH 1 and$L_2 = \{5, 6\}$ is the neighbor list of CH 2. Then a replacement is done by$((1,3),(1,5),(2,5),(2,6))$.

3.3.6. Selections of fault free routing path

The CH selects the fault-free routing path from the sensor node to BS using the GA algorithm. The best data among the clusters are transmitted as a new message to every CHs. Only those CHs free from fault can use their energy for data forwarding. A faulty CH cannot transmit the data due to low battery power.

For selecting fault free routing path, initially representing the placement of fault free nodes. Then path with the highest$L_f$is taken as a fault-free routing path. In this work, the self-healing process is used if there is any network connectivity issue for data forwarding. Assistant or human monitoring are the two categories of self-healing. When the network identifies a network failure, then it adopts a self-healing policy that removes the link that created the problem. With the help of the wake-up strategy, it wakes the faulty node. Hence, connectivity problem is avoided by the self-healing process. This fault tolerant routing can be applied in the field of the border surveillance system. This application is implemented with H-WSN. Sensor nodes are placed in the border area. After using the FTGSO algorithm, sensor nodes sense the intrusion activity in the border area and give the sensed data to the BS. Then BS decide whether to take action or not.

Fig. 4 represents the workflow of FTGSO. Proposed MO-GSO have 2 phases. On the first phase of population, a total number of nodes in the network is fixed using PSO and a selected number of clusters using the fittest value (F). Then CH is selected on the basis of multi-objectives. The second stage uses the GA algorithm. In this stage, faulty SN is selected and considered in all the clusters for fault tolerance. Finally, a fault free routing path is generated for data transmission. With the help of MO-GSO, fault occurrence is eliminated and increases the efficiency of the network. The algorithm is terminated when the fault free routing path is

Table 2
Notaion and meaning of the symbols.

Notations	Meaning
N_{nrm}	Total number of normal nodes
N_{int}	Total number of intermediate nodes
N_{adv}	Total number of advanced nodes
m_i	Portion of intermediate nodes
m_a	Portion of advanced nodes
n_o	Number of nodes
γ	Factor of intermediate nodes
β	Factor of advanced nodes
E_{nrm}	Energy of normal nodes
E_{int}	Energy of intermediate nodes
E_{adv}	Energy of advanced nodes
E_e	Initial energy
D	Distance between receiver and sender
E_c	Energy consumption
E_f	Transmitter
E_m	Amplifier
j	Number of packets
E_{ag}	Energy needed for one-bit aggregation
m	Number of messages
p_{in}^x	Position of each particle
ve_{in}^x	Velocity of each particle
x	Dimension
y_{in}^x	Previous best solution
m_n^x	Global best solution
$h_1\ , h_2$	Cognitive and social constant
SN_t	Number of sensor nodes
$N_n(x,y)$	Distance between nodes
dis_{avg}^2	Node average distance
d_r^2	neighbor average distance
E_{sn}	SN total energy
E_{co}	Energy used for data collection
E_{trans}	Data transmission
$R(X_y)$	The node radius
$R_{initial}$	Initial energy of CH
$R_{\max}$	The maximum amount of energy used by CH

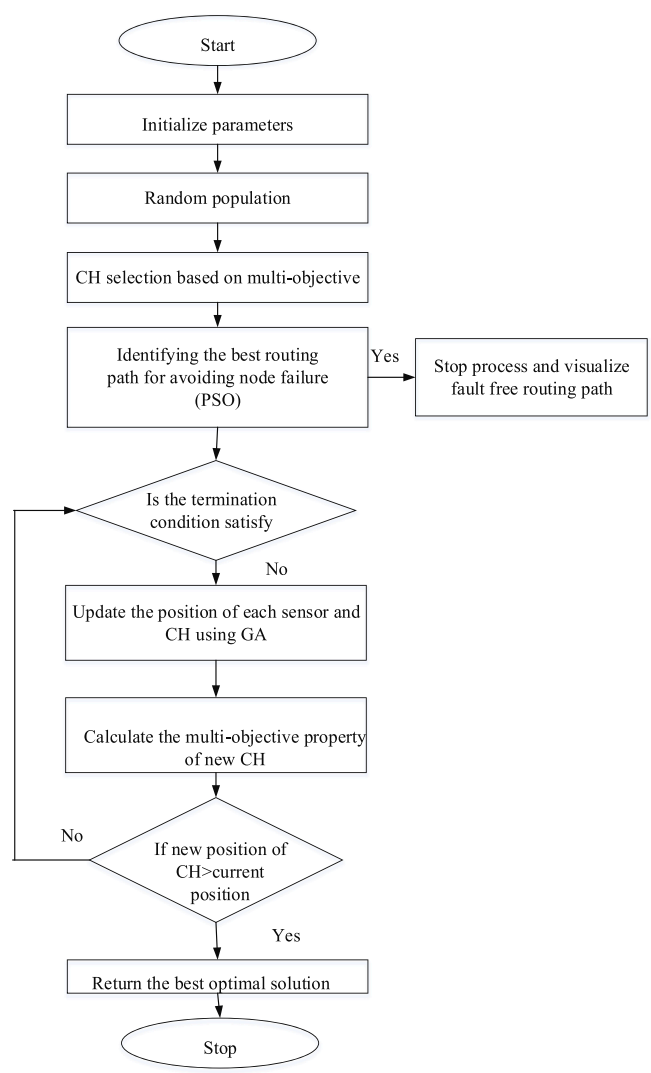

Fig. 4. flowchart of FTGSO.

obtained for data transmissions. The fault free routing path is determined based on the fitness evaluations for each sensor and CH in the path. A system that does not work properly employs the self-healing method that helps return to its working condition. Intrusion activity in the border area is sensed by the sensor node and gives BS data to decide which action to take.

Fig. 5 represents the application of the proposed method for a border surveillance system. Assume that a border surveillance system is developed in 2D heterogeneous WSN. For his application, the SNs are placed randomly on the border area. If there is an intrusion detected by the sensor node, it gives information to BS. The military control panels decide about the prevention action for intrusion.

4. Result and analysis

The entire implementations of the proposed work is carried out in the Matlab platform version 2021. In this work, single sink 2D architecture with a network area of $100m \times 100m$is used and there are 100–500 SNs randomly deployed. . The initial energy of the node is$2J$. 1024 bytes is the packet size, and different types of clusters are used. The location of the BS is [50,125]. The range of nodes lies between 40 and 50. The maximum number of bits which can be transferred between nodes is 1 Mbps, and the maximum number of rounds is 4000. A summary of these parameters is presented in Table 3.

4.1. Performance metrics

the following parameters are considered for evaluation: E2E, energy efficiency, PDR, PLR, and energy consumption.

PLR: It is the portion of the total lost of packets to the entire packets delivered from SN to sink node.

$$Packet\ Loss\ Ratio = \frac{Number\ of\ packets\ lost}{Total\ number\ of\ delivered\ packets} \tag{27}$$

Energy Consumption: It is the total amount of energy utilized between sensor nodes and sinks nodes during successful communication. Total energy consumption is denoted by Joule (J).

$$Energy\ Consumption = \frac{E_{Consumption}}{\eta \times D_{received}} \tag{28}$$

where$E_{Consumption}$ is the energy consumption of the total network and node total range denoted byη.$D_{received}$ is the total range of data packets received at the destination.

PDR: It is the ratio of the total packets reached at the sink node to the total packets generated at the SN.

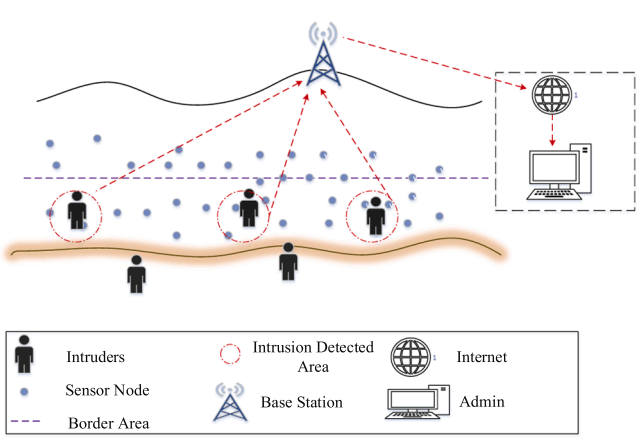

Fig. 5. Application of FTGSO- border surveillance system.

Table 3
Network Parameters.

Parameters	Value
Area of deployment	$100m \times 100m$
Number of Sink Node	1
Initial Energy of Nodes	$2J$
Number of nodes	100–500
Size of data packets	1024 bytes
Location of BS	[50, 125]
Number of Simulation run	18,000
Total rounds	2000
Energy consumption on the process of data aggregation	$2J$
h_1	2
h_2	2
Total iteration	100
Clusters	Different Type
Nodes Range	40–50
Maximum Network Throughput	$1\,Mbps$
Maximum Number of Rounds	4000
Threshold Distance	$70\,m$
Energy utilized by the amplifier for transmitting at a longer distance	$0.0013pJ/bit/m^4$
Energy used by the amplifier for transmitting at a shorter distance	$10\,pJ/bit/m^2$
Energy utilized by the electronic circuit amplifier for transmitting or receiving the signal	50nJ/bit
Data Aggregation Energy	5nj/bit

$$Packet\ Delivery\ Ratio = \frac{Total\ amount\ of\ packets\ reached\ at\ the\ \text{sink}\ node}{Total\ amunt\ of\ packet\ generated\ at\ the\ sensor\ node} \tag{29}$$

E2E: It is the amount of time taken for the data packet transmission across the network.

$$End - to - End\ Delay = \sum_{g=1}^{D_{nr}} \frac{(t_{nr} - t_{nt})}{V} \tag{30}$$

Time taken by the source node for generating data packets is t_{nt}, and the time taken by BS for receiving the data packets is t_{nr}.

Energy Efficiency: It is the minimum amount of energy used to perform a specific task.

$$Energy\ efficiency = \frac{total\ amount\ of\ output\ enery}{total\ amount\ of\ input\ enery} \tag{31}$$

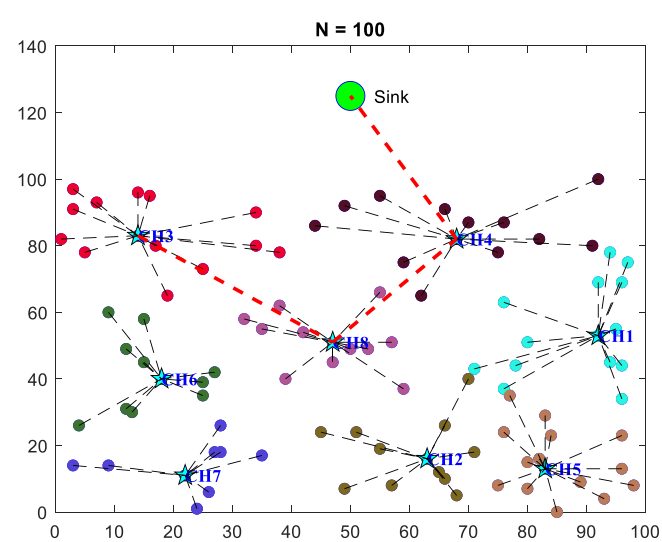

Fig. 6. Proposed FTGSO representation for $N = 100$.

4.2. Simulation results

The proposed FTGSO representation for different values of N is shown in Figs. 6-10. With the help of multi-objective analysis, the CH is selected from the clusters. After selecting CH from each cluster, a fault free routing path is provided with the help of GSO. Hence, data packets are transmitted from source to destination.

4.3. Performance comparison

Our proposed FTGSO method is compared with MCH-EOR, GA, PSO, GWO and ALO [27] in terms of the PLR, energy efficiency, PDR, E2E, energy consumption, analyzed for different values of density of nodes and node number. Results are represented below figures separately.

The E2E performance analysis with various optimization techniques is shown in Fig. 8. The end-to-end delay of existing methods is greater than the proposed scheme. When the number of SNs are increased, the E2E also increases. The proposed method utilizes 30 ms of E2E with respect to 500 nodes, and GA, ALO, PSO, GWO and MCH-EOR utilized120, 130,140, 70 and 40 ms of E2E for 500 nodes.

The comparison of the PLR of different approaches is indicated in Fig. 9. The proposed method gives a minimum packet loss ratio compared to the existing optimization routing algorithms. From Fig. 9, it can be observed that the PSO routing technique gives the highest PLR (9). This result is achieved due to proper CH selection, while the other optimization obtained poor results due to improper CH selection. The introduced method uses a 3.2% packet loss ratio to 500 nodes, and ALO, GA, MCH-EOR, GWO and PSO utilized 6.9, 6, 3.5, 4.8 and 9 percentage of packet loss ratios.

Fig. 10 represents the packet delivery ratio comparison. From Fig. 10, it is evident that the proposed method gives the highest PDR, and lower performance is given by PSO. This is due to the fitness function, which is good in selecting proper CHs. packet delivery ratio is decreased for all optimized routing techniques when the density of nodes is increased. The PDR of GA, ALO, PSO, GWO and MCH-EOR in the case of 500 nodes is 93, 92.2, 91, 94.2 and 95.5, respectively.

The energy consumption analysis of the proposed method with different routing models is introduced in Fig. 11. The energy consumption of the entire process is computed for 500 nodes. The proposed method gives better energy consumption performance than other existing routing methods. The fitness function generated by FTGSO has achieved this performance. This implies that the proposed method consumes a minimum amount of energy for data transmission from source to sink node. i.e. 0.19 J energy is used by the proposed system.

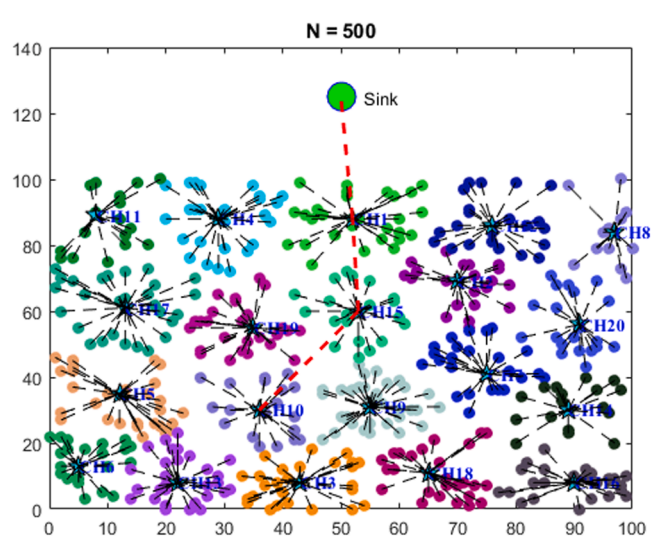

Fig. 7. : Proposed FTGSO representation for $N = 500$.

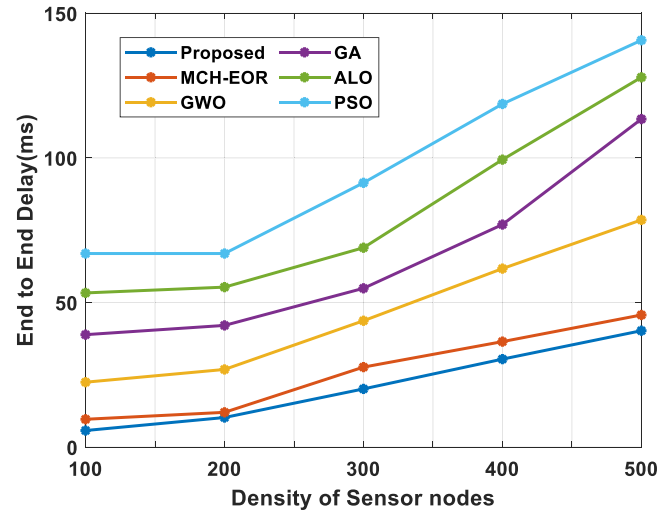

Fig. 8. Comparison of E2E with various optimization routing techniques.

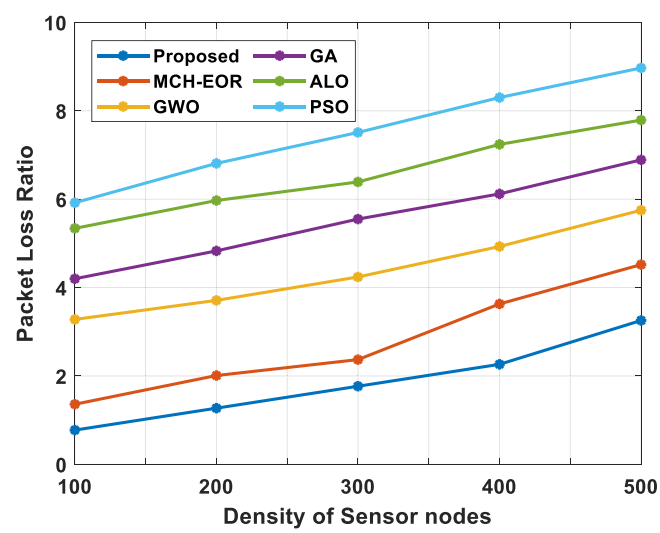

Fig. 9. Comparison of PLR with various optimization routing techniques.

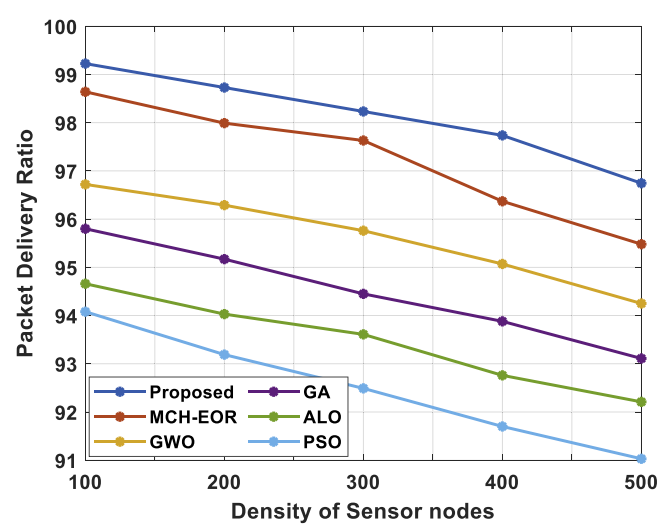

Fig. 10. Comparison of PDR with various optimization routing techniques.

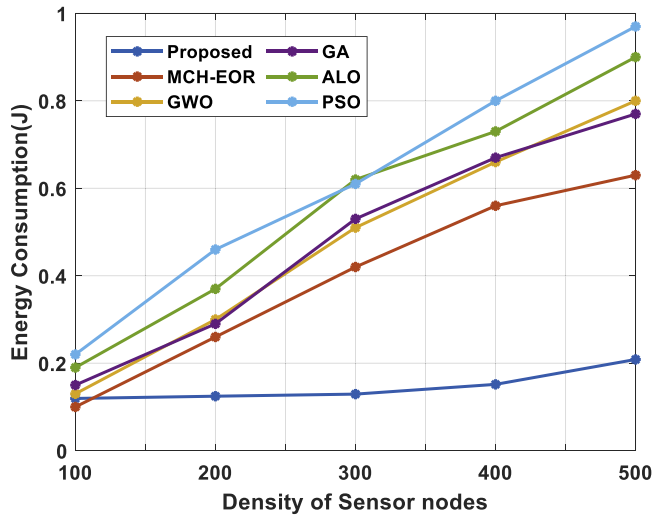

Fig. 11. Comparison of Energy Consumption with various optimization routing techniques.

The PSO method has the highest energy consumption (0.99 J) than other routing techniques.

The PDR of the proposed method with a different number of rounds is depicted in Fig. 12. The different number of nodes ($N = 100$, 200, 300, 400 and 500) are used to analyze the PDR of the proposed system. When the number of nodes is 100, the proposed system utilizes a packet delivery ratio below 10,000 rounds. From Fig. 12, it is observed that when the number of nodes increases and its round performances also increase.

Fig. 13 represents the proposed packet loss ratio analysis method for different values of N. When several nodes are 100, it gives a minimum value of round performance than other nodes. For $N = 200$, 300, 400 and 500 the packet loss ratio gives a round performance between 8000 and 1000. When the number of nodes increases, the round based PLR also increases.

The energy efficiency of the proposed system is given in Fig. 14. When the number of nodes increases, the energy efficiency also increased. From Fig. 14, it is evident that energy efficiency is higher when the number of nodes is 100. When the number of nodes is between 300 and 500 range, the energy efficiency performance for all round is almost similar.

Fig. 15 indicates the E2E performance of the proposed system with a various number of rounds. Fig. 15 depicts the increase in E2E with an

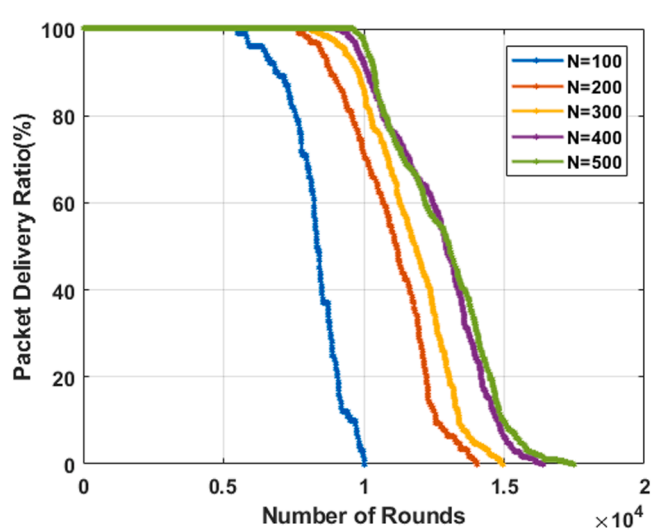

Fig. 12. Proposed Method PDR with different number of rounds.

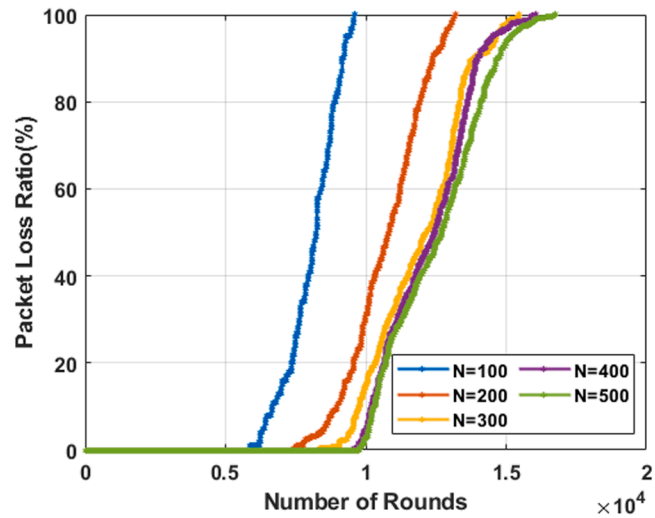

Fig. 13. Proposed Method Packet Loss Ratio with different number of rounds.

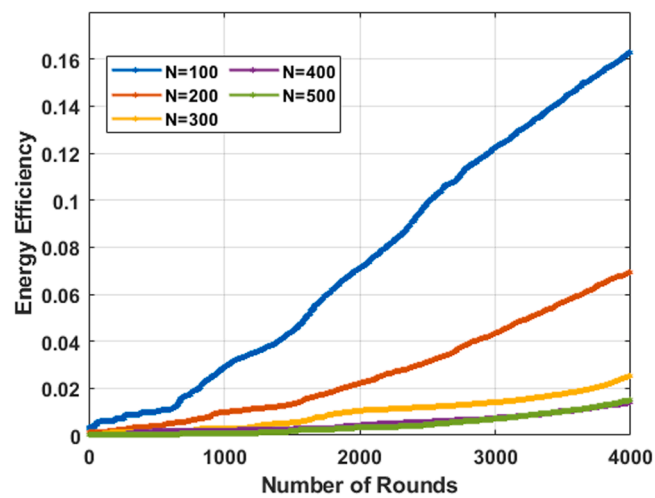

Fig.14. Proposed Method Energy Efficiency with different number of rounds.

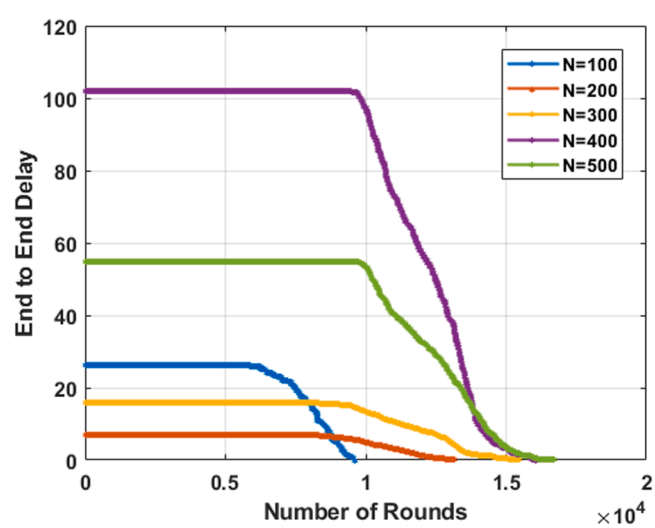

Fig.15. : Proposed Method E2E with different number of rounds.

increased number of nodes. After 10,000 rounds, the E2E delay gradually reduced for all values of N. Larger E2E is observed when the number of nodes is 500 and lower values of E2E delay are at $N = 100$.

4.4. Cost analysis

The computational cost of the algorithm is one of the important evaluation metric that provides information regarding the efficiency in data transmissions. The total cost of the network for data transmissions should be reduced to prove the performance efficiency of the proposed method compared to the existing methodologies. A comparison for cost for the proposed and existing methodologies is presented in Table 4.

Based on the comparison provided in the table, it is clear that the proposed method is more effective in transmitting the data compared to the existing methods. The cost values are determined for a total transmission rate of 50 data packets/seconds. Since the proposed method is fault tolerant and is capable of optimally finding the fault free routing path, the cost required in data transmission is reduced. The overall cost taken by the proposed method in transmitting 50 packets/seconds is 22.13 s whereas, the cost taken by MCH-EOR, GWO, GA, ALO and PSO are 56.35 s, 50.10 s, 46.87 s, 39.94 s and 30.81 s respectively.

Table 4
Cost analysis for the proposed and existing methods.

Methods	Cost (s)
MCH-EOR	56.35
GWO	50.10
GA	46.87
ALO	39.94
PSO	30.81
Proposed	**22.13**

4.5. Discussion

Upon simulating the proposed model, it has been identified that the proposed model is capable of providing effective fault tolerant routing thereby enabling effective data transmissions. The proposed model has been compared with several methods and algorithms to prove its performance efficacy. Moreover, the cost analysis performed proved its efficiency in transmitting the data packets throughout the network. On comparing the proposed scheme with the existing methodologies, the performance improvement of the proposed scheme can be identified. The scheme for load balancing introduced by Shahbaz et al. [20] resutled in enhanced scalability but, the routes discovered for data transmissions are not evaluated as network traffic may occur. Another priority based scheme was introduced for congestion control in the network by Hatamian et al. [21] that prioritized the data to idnetify the affordable routes. But, the quality of the data being prioritized in the scheme is not analyzed. The scheme presented by Naghibi and Barati [22] provided effective data aggregation by avoiding retransmissions of additional packets. But this method compromised with the dynamicity of the network. Also, the method presented by Yousefpoor et al. [23] optimized the large-scale WSNs by optimally routing the data packets but the security of the system is still under question. The scheme introduced by Sharifi and Barati [24] reduced the latency in data transmissions and enabled proper communication. But, this method still require deep analysis in the efficiency and computational cost of the system. Compared to the analyzed existing methodologies, the proposed method effectively overcame the existing difficulties and resulted in optimal outcomes.

5. Conclusion

In this paper, an optimal fault tolerant genetical swarm optimization (FTGSO) method is introduced with the help of multi-objectives for WSN. The FTGSO method helps to overcome the problem of network partitioning and network connectivity. In the FTGSO, data transmission is carried out by optimal routing path identification, and the performances are monitored by PDR, energy consumption, PLR, energy efficiency and end-to-end delay. These performance metrics are compared with other existing routing methods such as ALO, GA, GWO, MCH-EOR and PSO. Simulation results verify the performance of the proposed model with a higher packet delivery ratio of 96.8%, lower energy consumption of 0.19 J with a minimum packet loss ratio of 3.2% and 30 ms of E2E compared to other existing routing protocols. Also, the performance of the proposed method is compared in terms of E2E, energy efficiency, PLR and PDR. In future, the work will be extended to analyze the necessity of QoS challenges in real time applications. Then, instead of using single BS, the work will be extended to have multi-BS for achieving better performace. Further, this model can be analyzed for real time application with high density of WSN. In this work, only permananet failures of CHs is considered. In the future, energy aware routing will be designed for transient and partial failures of nodes and it can be used for several applications like surveillance and monitoring.

Author statement

We declare that this manuscript is original, has not been published before and is not currently being considered for publication elsewhere. We confirm that the manuscript has been read and approved by all named authors.

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All authors have equal contributions in this work.

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This article does not contain any studies with human participants or animals performed by any of the authors.

Consent to participate

All the authors involved have agreed to participate in this submitted article.

Consent to publish

All the authors involved in this manuscript give full consent for publication of this submitted article.

Data availability statement

Data sharing not applicable to this article.

Declaration of Competing Interest

Authors declare that they have no conflict of interest.

Data Availability

The authors do not have permission to share data.

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